

AI & Machine Learning – Task 2

Feature Engineering, Model Optimization & Performance Comparison

1. Objective

This report documents the methodology, implementation, and results of Task 2, which builds upon Task 1 by introducing feature engineering, feature scaling, multi-model training, and data-driven model selection on the California Housing dataset.

2. Dataset

The **California Housing Dataset** (20,640 samples, 8 input features) was used. The target variable is *Median House Value* (HousePrice). Input features include MedInc, HouseAge, AveRooms, AveBedrms, Population, AveOccup, Latitude, and Longitude.

3. Methodology

Step 1 – Import Libraries

pandas, NumPy, matplotlib, and scikit-learn modules (StandardScaler, LinearRegression, Ridge, DecisionTreeRegressor, train_test_split, metrics).

Step 2 – Load Dataset

Fetches the California Housing dataset and combined features with the target column into a single DataFrame.

Step 3 – Feature/Target Separation

Input matrix X (8 features) and target vector y (HousePrice) were separated to prepare for model training.

Step 4 – Feature Scaling (Critical)

StandardScaler was applied to X to normalize all features to mean=0 and std=1. This ensures no single feature dominates due to magnitude differences.

Step 5 – Train-Test Split

Data was split 80/20 (train/test) using random_state=42 for reproducibility. This evaluates models on unseen data to prevent data leakage.

Step 6 – Multi-Model Training

Three models were trained: (a) Linear Regression as a baseline, (b) Ridge Regression with alpha=1.0 to reduce overfitting via L2 regularization, and (c) Decision Tree Regressor with max_depth=5 to capture non-linear patterns.

Step 7 – Evaluation

Each model was evaluated using RMSE (Root Mean Squared Error) and R^2 Score. Lower RMSE and higher R^2 indicate better performance.

Step 8 – Visualization

An Actual vs Predicted scatter plot was generated for the best model, along with a grouped bar chart comparing all models on both metrics.

4. Results

Model	RMSE	R ² Score	Notes
Linear Regression	0.7256	0.7390	Baseline – linear relationships only
Ridge Regression	0.7256	0.7390	Same as LR; no multicollinearity benefit
Decision Tree	0.4055	0.9185	Best – captures non-linear patterns

5. Conclusion & Model Selection

The **Decision Tree Regressor (max_depth=5)** was selected as the best model. It achieved the highest R² score of 0.9185, meaning it explains approximately 92% of the variance in house prices. Its RMSE of 0.4055 is significantly lower than the linear models (0.7256), demonstrating strong predictive accuracy.

Linear Regression and Ridge Regression performed identically because the dataset does not suffer from significant multicollinearity, making regularization unnecessary in this case. The Decision Tree's ability to model non-linear feature interactions explains its superior performance.

6. Key Learnings

- Feature scaling is essential for distance-sensitive algorithms like Ridge.
- No single algorithm is universally best — comparison is critical.
- R² and RMSE together provide a complete picture of model quality.
- Train-test splitting prevents overestimated accuracy from data memorization.
- A structured ML pipeline (load → preprocess → train → evaluate → compare) mirrors industry practice.

Appendix: Model Performance Charts

Task 2 - Model Performance Comparison

